

AI Anticipates the Asking Price



School of Technology & Computing





Agenda

- Goal, Purpose & Approach
- Background & Related Work
- Cloud Services
- AI Process Architecture
- Azure AutoML
- Technical Architecture
- Model Selection
- Visualizations/Metrics
- Front End & API
- Observations & Conclusion
- Future Potential Work
- Key References



Goal, Motivation, & Use



Empower property buyers



ID investment/sale opportunities



Housing price predictions



Approach



Research
Models &
Services



Review
Relevant
Literature



Analyze
Similar
Projects



Manual
Build & Test
Prototype(s)



Azure
Auto ML
& Metrics





Background



AI-driven pricing hazards



Random Forest and regression analysis



Amazon versus Azure cloud services



Related Work

★ Limit access to AI price estimations to mitigate hyper-inflation!

(2024) Modeling the Feedback of AI Price Estimations on Actual Market Values

<https://doi.org/10.48550/arxiv.2405.18434>

★ Random Forest model significantly improves predictions!

(2023) ML for Rental Price Prediction: Regression Techniques and Random Forest Model

<https://doi.org/10.2139/ssrn.4587725>

★ Make better-informed sales and investment decisions!

(2024) Housing Market Intelligence: Data Science for Rental Price Forecasting

<https://doi.org/10.1109/INOCON60754.2024.10511646>



Cloud Services





AI Process Architecture

Data

Acquire

Prepare

Reduce

Model

Type

Build

Train

Test

Interaction

Application

API

Results



Front End & API

Feature Input

https://missvz.pythonanywhere.com

Enter Property Details

Number of Bathrooms:
4

Living Area (Sq Ft):
2290

Lot Size (Sq Ft):
13416

Waterfront Property:
☐ Yes ☒ No

View Rating (0-10):
0

Condition Rating (1-10):
4

Grade Rating (1-20):
9

Above Ground Area (Sq Ft):
2290

Basement Area (Sq Ft):
0

Year Built:
1981

Year Renovated:
0

Zip Code:
98007

Latitude:
47.6

Longitude:
-122.1

Living Area of 15 Nearest Neighbors (Sq Ft):
2680

Lot Size of 15 Nearest Neighbors (Sq Ft):
13685

Submit

- Demonstration
- HTML/CSS GUI
- Flask app

Prediction Result

\$727,477.59

Go Back

6:46

pythonanywhere.com

Lot Size (Sq Ft):

Waterfront Property:
☐ Yes ☒ No

View Rating (0-10):

Condition Rating (1-10):

Grade Rating (1-20):

Above Ground Area (Sq Ft):

Basement Area (Sq Ft):

Year Built:

Year Renovated:

Zip Code:

Latitude:

Longitude:

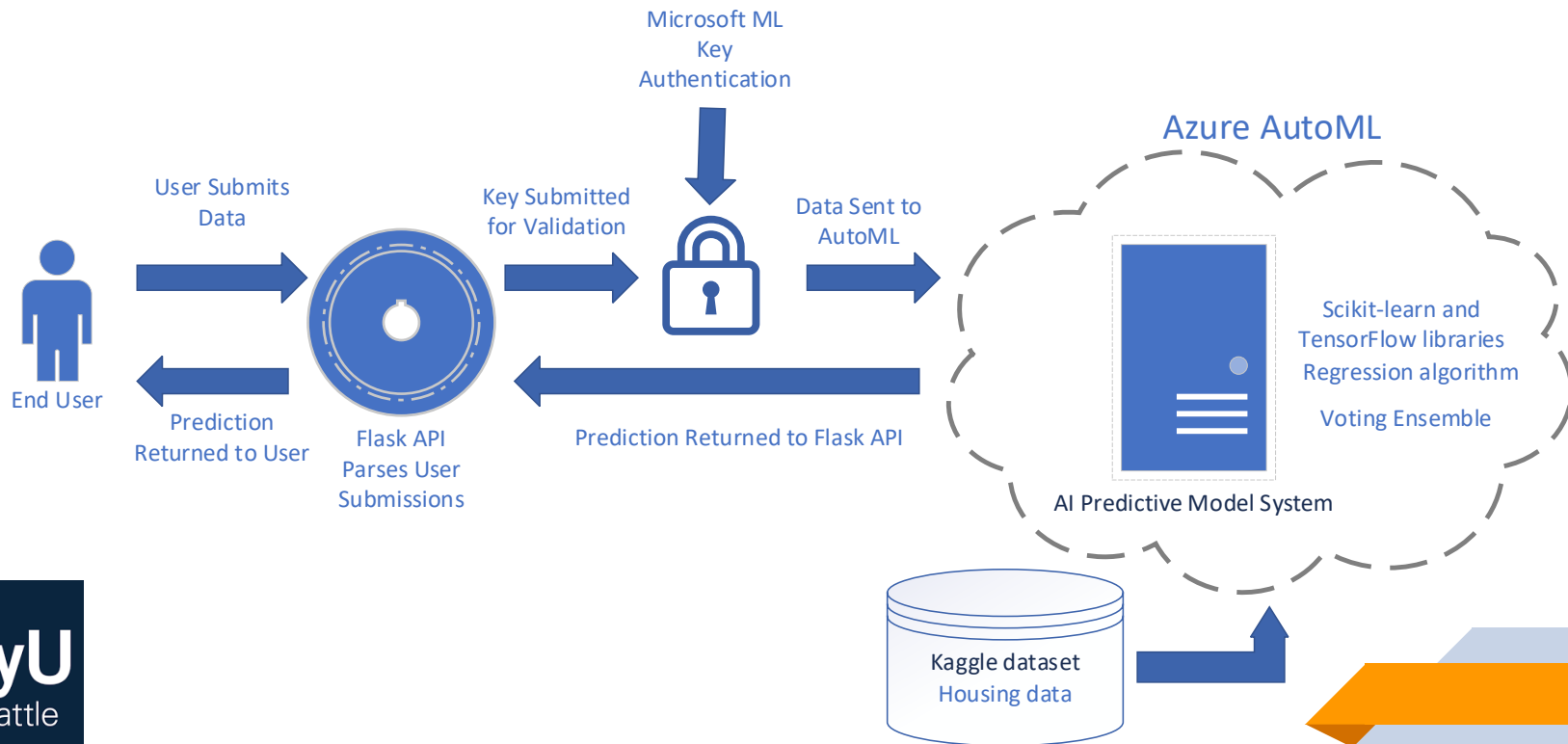
Living Area of 15 Nearest Neighbors (Sq Ft):

Lot Size of 15 Nearest Neighbors (Sq Ft):

Submit

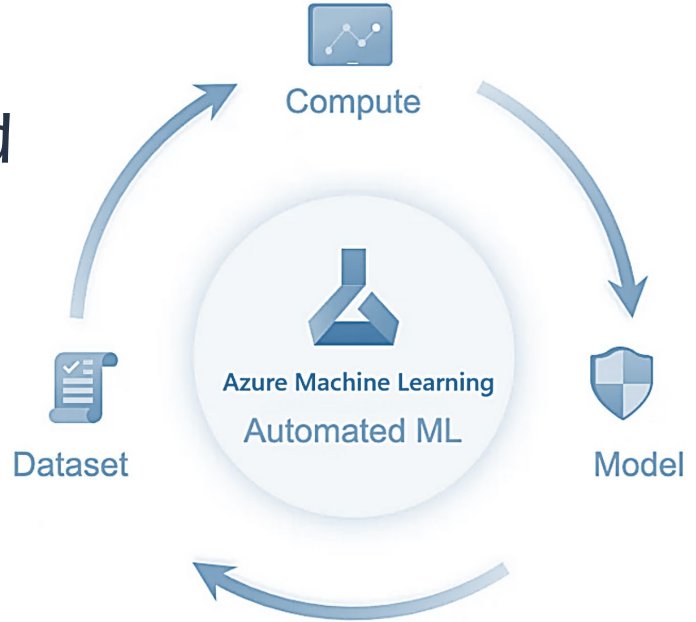
Mobile-friendly

Technical Architecture



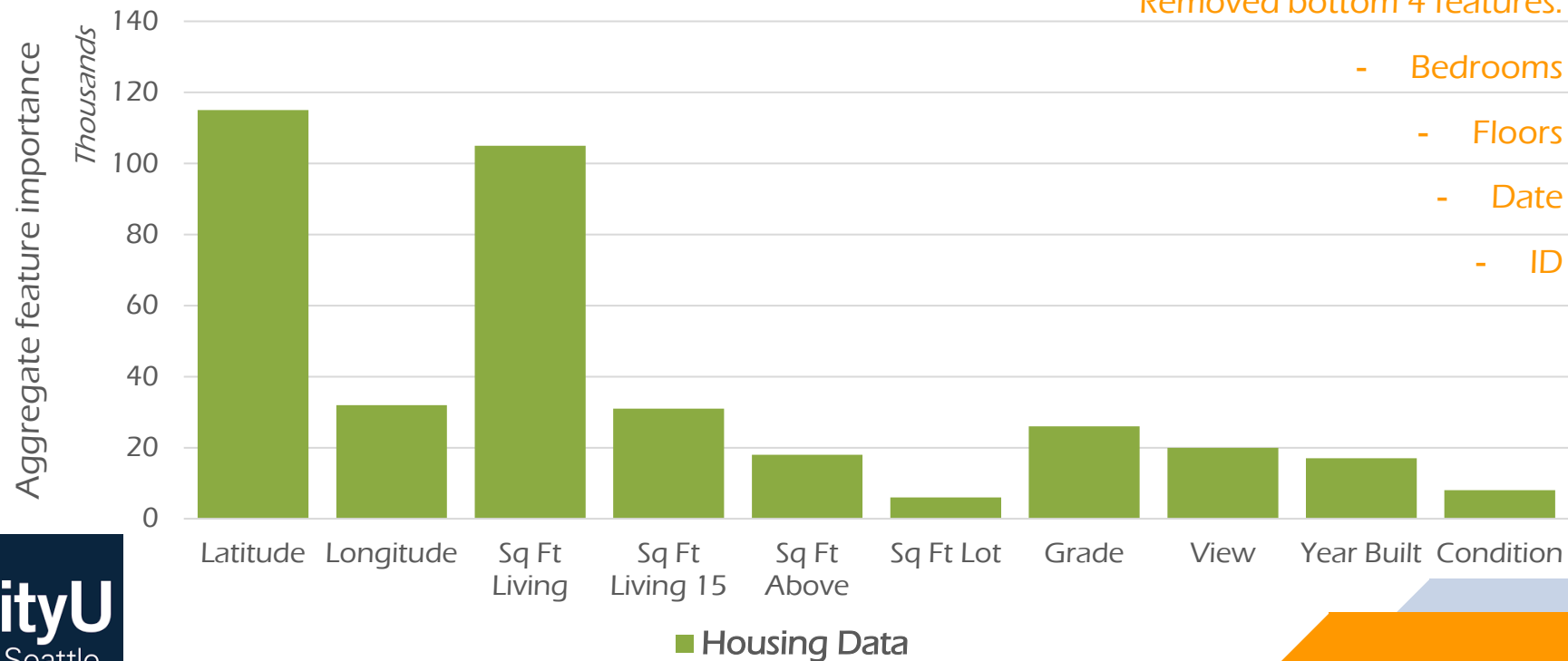
Microsoft's AutoML

- No Deep ML expertise needed
- Scalable with Azure AI Tools
- Accelerates Model Building
- High-Performance Models

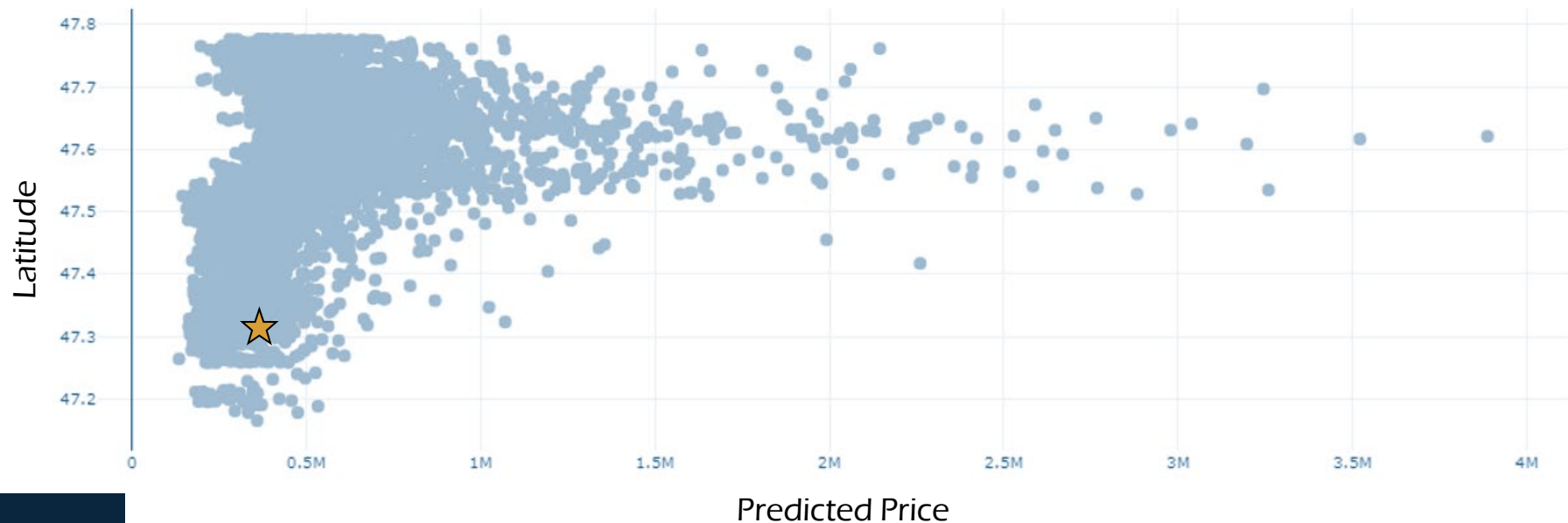




TOP 10 Relevant Features



Most Relevant Feature



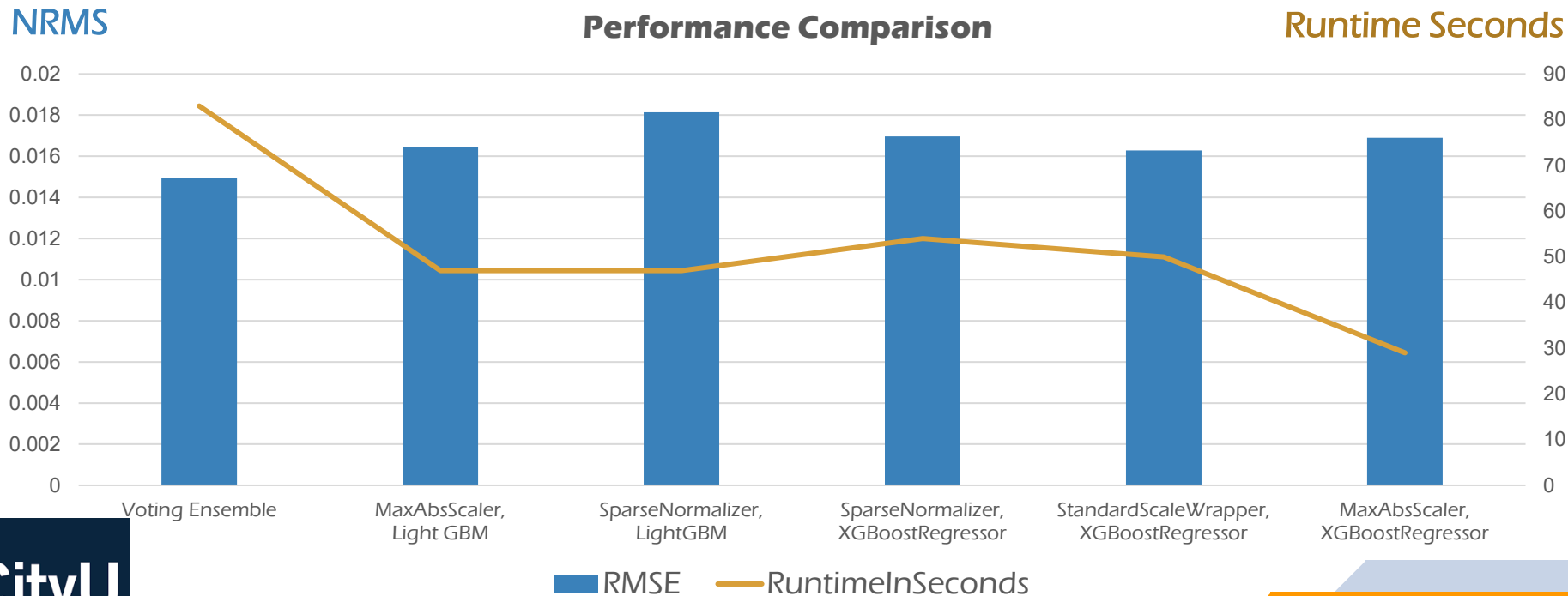


BEST Regression Models

1. Voting Ensemble (0.01494) - winner
2. MaxAbsScaler, Light GBM
3. SparseNormalizer, LightGBM
4. SparseNormalizer, XGBoostRegressor
5. StandardScaleWrapper, XGBoostRegressor
6. MaxAbsScaler, XGBoostRegressor

LOWEST
Normalized
Root Mean
Squared Error
< 0.02

Metrics





Metrics captured

Explained variance	0.90053
Mean absolute error	64229
Mean absolute percentage error	12.354
Median absolute error	38086
Normalized mean absolute error	0.0084235
Normalized median absolute error	0.0049949
Normalized root mean squared error	0.014941
Normalized root mean squared log error	0.035939
R2 score	0.90052
Root mean squared error	1.1393e+5
Root mean squared log error	0.16645
Spearman correlation	0.94997

Other
relevant metrics
captured by
Auto ML

Observations

Moderate
algorithm
prediction times

Azure AutoML
optimized speed &
scaling

SPEED



Key-based
authentication

Azure built-in
security &
encryption
protected data

SECURITY



Azure AutoML
straightforward
setup

Integrated cloud
services streamlined
deployment

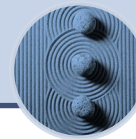
SETUP



Flask light for
API & web UI

Azure AutoML
abstracted complex
model building

SIMPLICITY





Conclusion

Why Auto ML was a better choice?

- Simplified automated model development
- Enabled rapid algorithm testing
- Efficient cloud processing
- Strong security (key-based auth & encrypt)
- Completed setup in 2-3 days as planned
- Delivered high-quality results (RMSE metrics)

Why was Flask chosen?

- Simple, flexible, and easily integrated with AI models



Future Potential Work

- Integration with CI/CD
- Expanding the Dataset
- Enhancing the User Interface
- User Feedback and Iterative Development



Key References

 Microsoft. (n.d.). Tutorial:
Train your first machine learning model - Automated ML.

 Sukhmandeep Singh Brar. (2024). Housing Price Dataset.
Kaggle.com. <https://www.kaggle.com/datasets/sukhmandeepsinghbrar/housing-price-dataset/code>



Verónica Elze
MS of AI



Apoorva Guru
MS in DS

choukrane
Баярлалаа gràcies
teşekkür
благодаря obrigado mulțumesc
dank u köszönöm Ďakujem
kiitos merci tack
terima kasih dhanṡavād
kop khun obrigada a dank tñnan
shukran dziękuję Xièxiè diolch
dēkujī תודה
Thank You
origatō paldies nandri Mh'gōi gracies
감사합니다 Հնրհալպալոթյունն спацибо
Дякую faleminderit ευχαριστώ
grazie ederim хвала takk
hvala ačiū ngiyabonga danke
Благодарам
grozzi tak dankie
maholo



Teagan Scharlau
DIT



George Stanley
DIT

Additional References

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<https://doi.org/10.48550/arxiv.2405.18434>